

The Mathemaniac

IIT Foundation & Olympiad Module for Class IX

Physics

Theme 1: Measurements and Experimentation

1. Units and measurements
2. Density and relative density
3. Experimental skills

Theme 2: Motion

1. Distance, displacement, speed and velocity
2. Acceleration
3. Graphs of motion

Theme 3: Force and Laws of Motion

1. Newton's laws
2. Momentum and inertia
3. Numerical applications

Theme 4: Fluids

1. Pressure in liquids and gases
2. Atmospheric pressure
3. Buoyancy basics

Theme 5: Heat

1. Temperature and heat
2. Expansion
3. Change of state

Theme 6: Light and Sound

1. Reflection and refraction
2. Mirrors and lenses
3. Sound waves and echo

Theme 7: Olympiad Enrichment

1. Advanced graph analysis
2. Multi-concept numericals
3. Logical physics reasoning